



NORTH SYDNEY BOYS HIGH SCHOOL

2015 HSC ASSESSMENT TASK 3

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Mr Lin
- ☐ Mr Weiss
- ☐ Ms Ziazaris
- ☐ Mr Zuber

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	Total	Total
Mark	$\frac{\quad}{10}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{70}$	$\frac{\quad}{100}$

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1 – 10

1. What is the value of

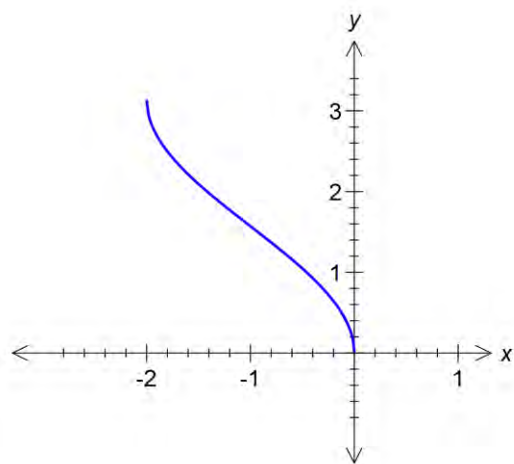
$$\lim_{x \rightarrow 0} \frac{\sin 4x}{5x} ?$$

- (A) 0
- (B) $\frac{4}{5}$
- (C) 1
- (D) $\frac{5}{4}$

2. $y = f(x)$ is a linear function with gradient $\frac{1}{4}$, find the gradient of $y = f^{-1}(x)$.

- (A) 4
- (B) $\frac{1}{4}$
- (C) -4
- (D) $-\frac{1}{4}$

3.



Which of the following best describes the above function?

(A) $y = \sin^{-1}(x + 1)$

(B) $y = \sin^{-1}(x) + 1$

(C) $y = \cos^{-1}(x + 1)$

(D) $y = \cos^{-1}(x) + 1$

4. What are the coordinates of the point that divides the interval joining the points A(-6,4) and B(-2,-10) externally in the ratio 1:3?

(A) (-8,8)

(B) (-8,11)

(C) (2,8)

(D) (2,11)

5. Which of the following is the solution to $\frac{2}{x-2} < 2$?

(A) $x < 2$ or $x > 3$

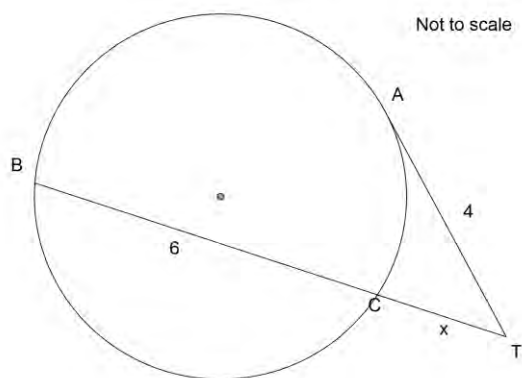
(B) $2 < x < 3$

(C) $-2 < x < 3$

(D) $-3 < x < 2$

6. The polynomial $P(x) = 2x^3 - 8x^2 + 7x - 14$ has roots α , $-\alpha$ and β . What is the value of β ?
- (A) 2
 (B) -2
 (C) 4
 (D) -4

7. The line TA is a tangent to the circle at A and TB is a secant meeting the circle at B and C.



Given that $TA = 4$, $CB = 6$ and $TC = x$, what is the value of x ?

- (A) 2
 (B) 4
 (C) 6
 (D) 8

8. Given that $\log_a 4 = x$, find an expression for $a^{\frac{3x}{2}}$

- (A) 2
 (B) 4
 (C) 8
 (D) 16

9. Find the gradient of the normal to the parabola $x = 6t$, $y = 3t^2$ at the point where $t = -2$.

- (A) -2
 (B) $-\frac{1}{2}$
 (C) $\frac{1}{2}$
 (D) 2

10. An approximate solution to the equation $f(x) = x + 2 \log_e x$ is $x = 0.5$. Using one application of Newton's method, a more accurate approximation is given by:

(A) $0.5 - \frac{0.5 + \log_e 0.25}{5}$

(B) $0.5 + \frac{0.5 + \log_e 0.25}{5}$

(C) $0.5 - \frac{5}{0.5 + \log_e 0.25}$

(D) $0.5 + \frac{5}{0.5 + \log_e 0.25}$

Section II

60 Marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question on a NEW page. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks) Start a NEW page.

- (a) When the polynomial $P(x) = 2x^3 - 3x^2 + ax - 2$ is divided by $(x + 1)$ the remainder is 7. What is the value of a ? 2
- (b) (i)
$$\int \frac{1}{(x + 4)^2} dx$$
 1
- (ii)
$$\int \frac{1}{x^2 + 4} dx$$
 1
- (iii)
$$\int \frac{x}{x^2 + 4} dx$$
 2
- (iv)
$$\int \frac{x}{(x^2 + 4)^2} dx$$
 2
- (c) Find the acute angle between the lines $y = 2x + 1$ and $2x + 5y - 2 = 0$ 2
- (d) Evaluate
$$\int_0^{\frac{\pi}{6}} \sin^2 2x \, dx$$
 3
- (e) Find the general solution to $2\cos^2 x = 1$ 2

Question 12 (15 Marks) Start a NEW page.

(a) (i) Without using calculus, sketch the graph of $P(x) = x(x + 2)(1 - x)^2$ 2

(ii) Hence solve $x(x + 2)(1 - x)^2 < 0$ 1

(b) Using the substitution $u = \frac{1}{x}$ find the exact value of:

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$$

3

(c) (i) A chef takes an onion tart out of the fridge at 4°C into a room where the air temperature is 25°C . The rate at which the onion tart warms follows Newton's law, that is:

$$\frac{dT}{dt} = -k(T - 25)$$

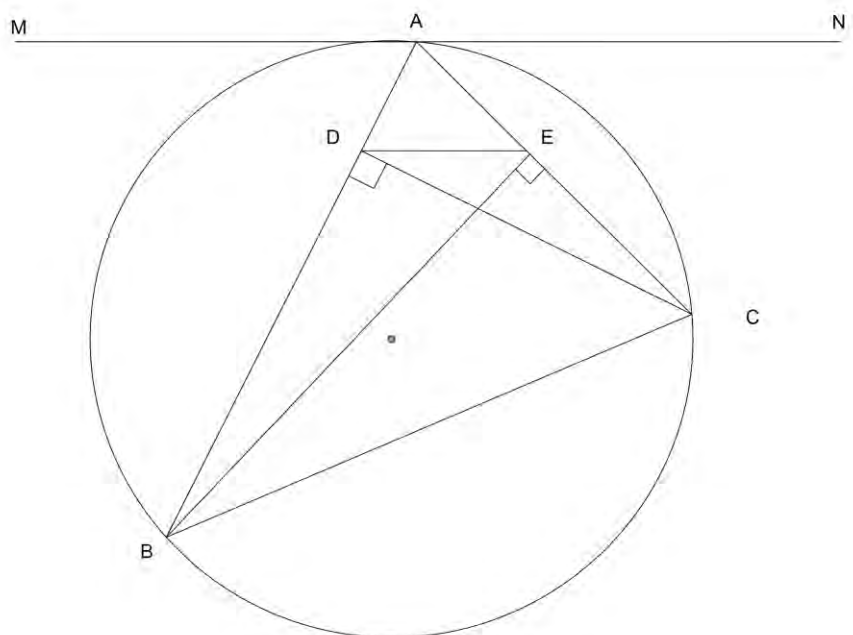
where k is a positive value, time t is measured in minutes and temperature T is measured in degrees Celsius.

Show that $T = 25 - Ae^{-kt}$ is a solution to $\frac{dT}{dt} = -k(T - 25)$ and find the value of A . 2

(ii) The temperature of the onion tart reaches 15°C in 45 minutes. Find the exact value of k . 2

(iii) Find the temperature of the onion tart 90 minutes after being removed from the fridge. 1

(d) (i)



ABC is a triangle inscribed in a circle. MAN is the tangent at A to the circle ABC . CD and BE are altitudes of the triangle.

Copy the diagram into your answer booklet.

- | | | |
|-------|--|----------|
| (ii) | Give a reason why $BCED$ is a cyclic quadrilateral | 1 |
| (iii) | Hence show that DE is parallel to MAN . | 3 |

End of Question 12

Question 13 (15 Marks) Start a NEW page

(a) Is the graph of $y = \log_8 x^2$ identical to $y = 2 \log_8 x$? Give a reason for your answer. 1

(b) (i) A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$ given by $a = x + \frac{3}{2}$. Initially the particle is 5m to the right of O and moving towards O with a speed of 6 ms^{-1} .

Explain whether the particle is initially speeding up or slowing down. 1

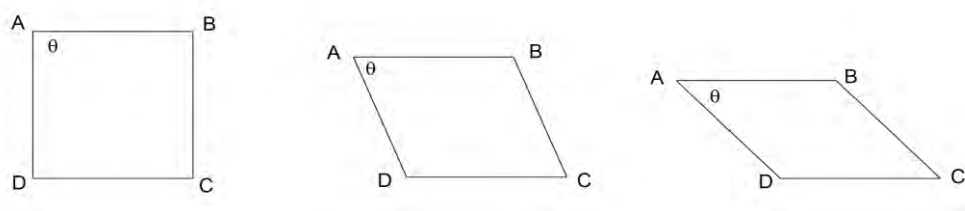
(ii) Find an expression for v^2 in terms of x . 2

(iii) Find where the particle changes direction. 1

(c) (i) Express $3 \cos \theta - \sqrt{3} \sin \theta$ in the form $A \cos(\theta + \alpha)$ 2

(ii) Hence, or otherwise, solve $3 \cos \theta - \sqrt{3} \sin \theta + 3 = 0$ for $0 \leq \theta \leq 2\pi$ 2

(d) (i)



A square $ABCD$ of side 1 unit is gradually 'pushed over' to become a rhombus. The angle at A (θ) decreases at a constant rate of 0.1 radian per second.

At what rate is the area of rhombus $ABCD$ decreasing when $\theta = \frac{\pi}{6}$? 3

(ii) At what rate is the shorter diagonal of the rhombus $ABCD$ decreasing when $\theta = \frac{\pi}{3}$ 3

Question 14 (15 Marks) Start a NEW page.

- (a) Prove that $11^{2n} + 11^n + 8$ is a multiple of 10 for all positive integers n 3
- (b) (i) Show that $\frac{d}{dx}(x \sin^{-1} x + \sqrt{1 - x^2}) = \sin^{-1} x$ 2
- (ii) Hence, using a similar expression, find a primitive for $\cos^{-1} x$ 1
- (iii) The curves $y = \sin^{-1} x$ and $y = \cos^{-1} x$ intersect at $P\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right)$.
The curve $y = \cos^{-1} x$ also intersects with the x axis at Q . 3
Find the area enclosed by the x -axis and the arcs OP and PQ .
- (c) (i) A parabola has parametric equations
 $x = t^2 + 1$
 $y = 2(2t + 1)$
Sketch the parabola showing its orientation and vertex. 1
- (ii) Point P is the point on the parabola where $t = p$
Point P' is the point on the parabola where $t = -p$
Find the equation of the locus of the midpoint of PP' and state its geometrical significance 2
- (iii) A line with gradient m passes through $(0, 5)$ and cuts the parabola at distinct points Q and R .
Find the range of possible values for m . 3

End of Examination.

Multiple Choice Answers

1./ B

2./ A

3./ C

4./ B

5./ A

6./ C

7./ A

8./ C

9./ C

10./ A

Question 11

Marks

(a) $P(x) = 2x^3 - 3x^2 + ax - 2$

$$P(-1) = 2(-1)^3 - 3(-1)^2 + a(-1) - 2$$

$$= -a - 7$$

$$\therefore 7 = -a - 7$$

$$\therefore a = -14$$

✓

✓

(b) (i) $\int \frac{1}{(x+4)^2} dx$

$$= \int (x+4)^{-2} dx$$

$$= (x+4)^{-1} + C$$

$$= \frac{-1}{x+4} + C$$

✓

(ii) $\int \frac{1}{x^2+4} dx$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) + C$$

✓

(iii) $\int \frac{x}{x^2+4} dx$

$$= \frac{1}{2} \int \frac{2x}{x^2+4}$$

✓

$$= \frac{1}{2} \log_e (x^2+4) + C$$

✓

$$\begin{aligned} \text{(iv)} \quad & \int \frac{x}{(x^2+4)^2} dx \\ &= \int x (x^2+4)^{-2} dx \\ &= \left(-\frac{1}{2}\right) (x^2+4)^{-1} + C \\ &= -\frac{1}{2(x^2+4)} + C \end{aligned}$$

✓✓

$$\begin{aligned} \text{(c)} \quad & y = 2x + 1 \\ \therefore m_1 &= 2 \end{aligned}$$

$$2x + 5y - 2 = 0$$

$$5y = -2x + 2$$

$$y = -\frac{2}{5}x + \frac{2}{5}$$

$$\therefore m_2 = -\frac{2}{5}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{2 - (-2/5)}{1 + (2)(-2/5)} \right|$$

$$= \left| \frac{2^{2/5}}{1/5} \right|$$

$$\therefore = |12|$$

$$\therefore \theta = 85^\circ 14' 11'' \text{ (nearest second)}$$

✓

✓

$$(d) \int_0^{\frac{\pi}{6}} \sin^2(2x) dx$$

$$= \int_0^{\frac{\pi}{6}} \left(\frac{1}{2} - \frac{1}{2} \cos 4x \right) dx$$

$$= \left[\frac{x}{2} - \frac{1}{8} \sin 4x \right]_0^{\frac{\pi}{6}}$$

$$= \left(\frac{\pi}{12} - \frac{1}{8} \sin \left(\frac{4\pi}{6} \right) \right) - \left(\frac{0}{2} - \frac{1}{8} \sin(4 \times 0) \right)$$

$$= \frac{\pi}{12} - \frac{1}{8} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{12} - \frac{\sqrt{3}}{16}$$

$$(e) 2 \cos^2 x = 1$$

$$\cos^2 x = \frac{1}{2}$$

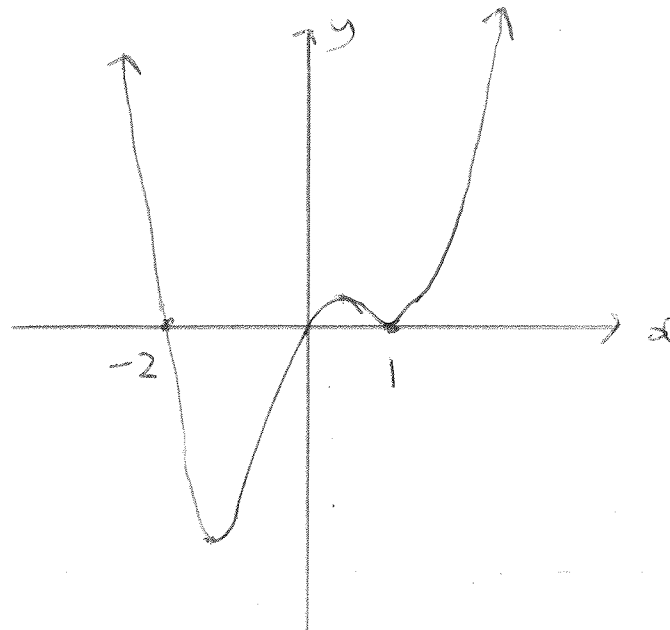
$$\cos x = \pm \frac{1}{\sqrt{2}}$$

$$\therefore x = n\pi \pm \frac{\pi}{4} \text{ (where } n \text{ is an integer)}$$

Question 12

Marks

(a)



✓✓

(i) $-2 < x < 0$

✓

(b) $u = \frac{1}{x}$

$$\therefore \frac{du}{dx} = -\frac{1}{x^2}$$

$$du = -\frac{1}{x^2} dx$$

✓

when $x = 2$

$$u = \frac{1}{2}$$

when $x = 1$

$$u = 1$$

✓

$$\begin{aligned} \int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx &= \int_{\frac{1}{2}}^1 (-e^u) du \\ &= \int_{\frac{1}{2}}^1 e^u \end{aligned}$$

$$\begin{aligned} &= [e^u]_{\frac{1}{2}}^1 \\ &= e - e^{\frac{1}{2}} \\ &= e - \sqrt{e} \end{aligned}$$

✓

(c) (i) $T = 25 - Ae^{-kt}$

$$\frac{dT}{dt} = kAe^{-kt}$$

$$= -k(-Ae^{-kt})$$

$$= -k(\underbrace{25 - Ae^{-kt}}_T - 25)$$

$$= -k(T - 25), \text{ as required}$$

✓

when $t = 0$ $T = 4$

$$4 = 25 - Ae^{-0k}$$

$$4 = 25 - A$$

$$A = 21$$

✓

(ii) when $t = 45$ $T = 15$

$$15 = 25 - 21e^{-45k}$$

✓

$$-10 = -21e^{-45k}$$

$$21e^{-45k} = 10$$

$$e^{-45k} = 10/21$$

$$-45k = \log_e(10/21)$$

$$\therefore k = -\frac{\log_e(10/21)}{45}$$

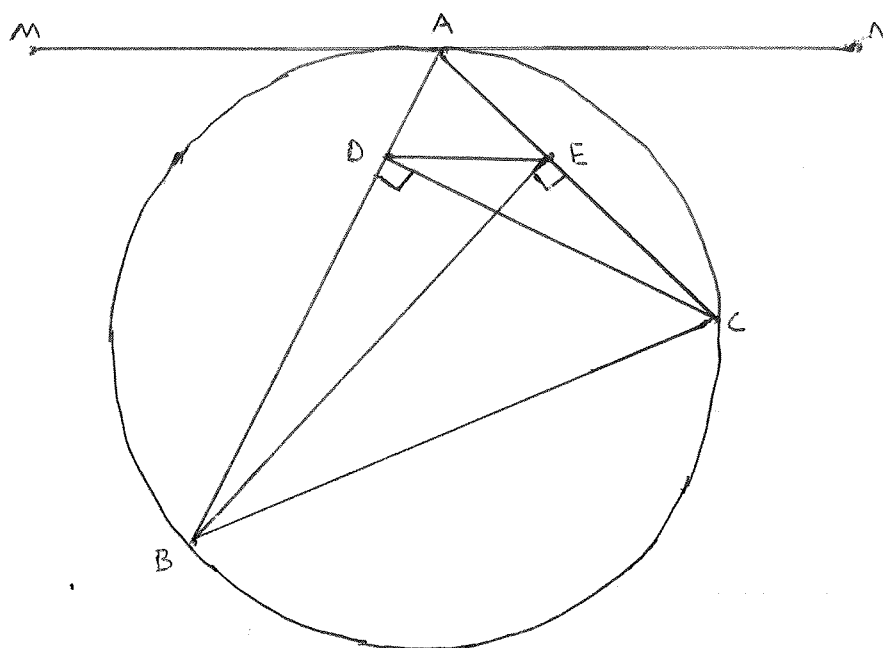
✓

$$(iii) T = 25 - 21e^{-90k}$$

$$= 20.238^\circ (3 \text{ d.p.})$$

✓

(d) (i)



(ii) BC subtends equal angles at D and E

✓

(iii) $\angle ABC = \angle AED$ (exterior angle of a cyclic quadrilateral is equal to the opposite interior angle)

✓

$\angle ABC = \angle NAC$ (angle between a chord and tangent is equal to the angle subtended by the chord at the circumference in the alternate segment)

✓

$\therefore \angle AED = \angle NAC$ (both equal $\angle ABC$)

$\therefore MAN \parallel DE$ (alternate angles are equal)

✓

Question 13

Marks

- (a) No. $y = \log_e x^2$ has domain all real x , $x \neq 0$ while $y = 2 \log_e x$ has domain $x > 0$.

✓

- (b) (i) It is slowing down since velocity is negative while acceleration is positive.

✓

(ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = a$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = x + \frac{3}{2}$$

$$\frac{d}{dx} v^2 = 2x + 3$$

$$v^2 = \int (2x + 3) dx$$

$$v^2 = x^2 + 3x + C$$

✓

when $x = 5$, $v = -6$

$$(-6)^2 = 5^2 + 3 \times 5 + C$$

$$36 = 25 + 15 + C$$

$$\therefore C = 36 - 40$$

$$C = -4$$

$$\therefore v^2 = x^2 + 3x - 4$$

✓

(iii) when $v^2 = 0$

$$\therefore x^2 + 3x - 4 = 0$$

$$(x+4)(x-1) = 0$$

$$\therefore x = -4 \text{ or } x = 1$$

$\therefore$ the particle changes direction at $x = 1$ (not at $x = -4$ since it turns back around at $x = 1$ and continues in the positive direction indefinitely) ✓

(c) (i) $A \cos(\theta + \alpha) = A \cos \theta \cos \alpha - A \sin \theta \sin \alpha$

$$\therefore A \cos \alpha = 3 \quad \dots \textcircled{1}$$

$$A \sin \alpha = \sqrt{3} \quad \dots \textcircled{2}$$

$$\therefore A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = (\sqrt{3})^2 + 3^2$$

$$\therefore A^2 (\sin^2 \alpha + \cos^2 \alpha) = 3 + 9$$

$$\therefore A^2 = 12$$

$$A = \sqrt{12} \text{ (take } A > 0)$$
 ✓

$$A = 2\sqrt{3}$$

sub into $\textcircled{2}$

$$\sqrt{12} \sin \alpha = \sqrt{3}$$

$$\sin \alpha = \frac{\sqrt{3}}{2\sqrt{3}}$$

$$\therefore \sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\therefore 3 \cos \theta - \sqrt{3} \sin \theta = \sqrt{12} \cos \left(\theta + \frac{\pi}{6} \right)$$

$$(ii) \quad 3 \cos \theta - \sqrt{3} \sin \theta + 3 = 0$$

$$\therefore 3 \cos \theta - \sqrt{3} \sin \theta = -3$$

$$\therefore \sqrt{12} \cos \left(\theta + \frac{\pi}{6} \right) = -3$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = -\frac{3}{\sqrt{12}}$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = -\frac{3 \times \sqrt{12}}{12}$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = -\frac{6\sqrt{3}}{12}$$

$$\therefore \cos \left(\theta + \frac{\pi}{6} \right) = -\frac{\sqrt{3}}{2}$$

$$\text{since } 0 \leq \theta \leq 2\pi, \quad \frac{\pi}{6} \leq \theta + \frac{\pi}{6} \leq \frac{13\pi}{6}$$

$$\therefore \theta + \frac{\pi}{6} = \frac{5\pi}{6} \text{ or } \frac{7\pi}{6}$$

$$\therefore \theta = \frac{4\pi}{6} \text{ or } \frac{6\pi}{6}$$

$$\therefore \theta = \frac{2\pi}{3} \text{ or } \pi$$

(d) (i) $A = 2 \times \frac{1}{2} \times 1 \times 1 \times \sin \theta$

(area of 2 congruent isosceles triangles)

$\therefore A = \sin \theta$

$\therefore \frac{dA}{d\theta} = \cos \theta$

$\frac{dA}{dt} = \frac{dA}{d\theta} \times \frac{d\theta}{dt}$

$= \cos\left(\frac{\pi}{6}\right) \times 0.1$

$= \frac{\sqrt{3}}{20} \text{ units}^2/\text{second}.$

(ii) let the shorter diagonal be l .

$l^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \times \cos \theta$

$l^2 = 2 - 2\cos \theta$

$\therefore l = \sqrt{2 - 2\cos \theta} \quad (l > 0)$

$\frac{dl}{d\theta} = 2 \sin \theta \times (2 - 2\cos \theta)^{-\frac{1}{2}} \times \frac{1}{2}$

$= \frac{\sin \theta}{\sqrt{2 - 2\cos \theta}}$

Marks.

$$\frac{dl}{dt} = \frac{dl}{d\theta} \times \frac{d\theta}{dt}$$

$$= \frac{\sin(\pi/3)}{\sqrt{2 - 2\cos\pi/3}} \times 0.1$$

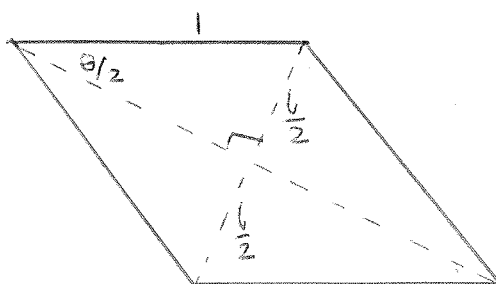
$$= \frac{\sqrt{3}/2}{\sqrt{2 - 2 \times (\frac{1}{2})}} \times 0.1$$

$$= \frac{\sqrt{3}/2}{1} \times 0.1$$

$$= \frac{\sqrt{3}}{20} \text{ units/second}$$

✓

alternative solution for (d)(ii)



since the diagonals of a rhombus meet at right angles

$$\sin\left(\frac{\theta}{2}\right) = \left(\frac{l}{2}\right)/1$$

$$\therefore l = 2 \sin \frac{\theta}{2}$$

$$\frac{dl}{d\theta} = 2 \times \frac{1}{2} \times \cos \frac{\theta}{2}$$

$$= \cos \frac{\theta}{2}$$

$$\frac{dl}{dt} = \frac{dl}{d\theta} \times \frac{d\theta}{dt}$$

$$= \cos\left(\frac{\pi/3}{2}\right) \times 0.1$$

$$= \sqrt{3}/2 \times 0.1$$

$$= \frac{\sqrt{3}}{20} \text{ units/second}$$

Question 14

Marks

(a) Base Case ($n=1$)

$$\begin{aligned} 11^{2 \times 1} + 11^1 + 8 &= 121 + 11 + 8 \\ &= 140 \\ &= 14 \times 10 \end{aligned}$$

which is divisible by 10

Assume true for $n=k$

i.e. assume $11^{2k} + 11^k + 8 = 10M$
where M is an integer

$$\therefore 11^{2k} = 10M - 11^k - 8 \quad \dots \textcircled{1}$$

Prove true for $n=k+1$

i.e. prove $11^{2(k+1)} + 11^{k+1} + 8$ is divisible by 10

$$11^{2(k+1)} + 11^{k+1} + 8 = 11^{2k+2} + 11^{k+1} + 8$$

$$= 11^2 \times 11^{2k} + 11^{k+1} + 8$$

$$= 121 \times (10M - 11^k - 8) + 11 \times 11^k + 8$$

(from $\textcircled{1}$)

$$= 1210M - 121 \times 11^k - 968 + 11 \times 11^k + 8$$

$$= 1210M - 110 \times 11^k - 960$$

$$= 10(121M - 11 \times 11^k - 96)$$

which is divisible by 10

∴ the proposition is true by the process of mathematical induction.

$$(b) (i) \frac{d}{dx} (x \sin^{-1}(x) + \sqrt{1-x^2})$$

$$= \left(\frac{d}{dx}(x) \right) \sin^{-1}(x) + x \left(\frac{d}{dx} \sin^{-1}(x) \right) + \left(\frac{1}{2} \right) (2x) (1-x^2)^{-\frac{1}{2}}$$

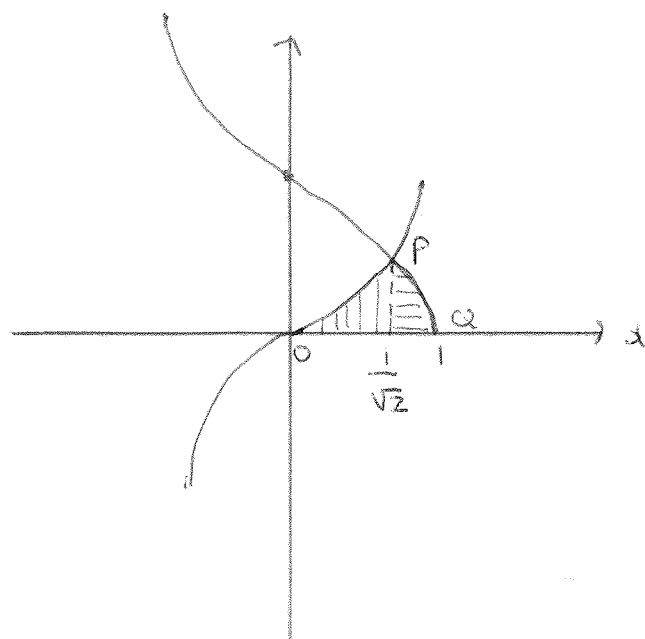
$$= 1 \cdot \sin^{-1}(x) + \frac{x}{\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}}$$

$$= \sin^{-1}(x) + \frac{x}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}}$$

$$= \sin^{-1}(x), \text{ as required}$$

$$(ii) x \cos^{-1}(x) - \sqrt{1-x^2}$$

(iii)



$$A = \int_{\frac{1}{\sqrt{2}}}^1 (\cos^{-1} x) dx + \int_0^{\frac{1}{\sqrt{2}}} \sin^{-1} x dx$$

✓

$$= \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_{\frac{1}{\sqrt{2}}}^1 + \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^{\frac{1}{\sqrt{2}}}$$

$$= (1 \times 0 - \sqrt{1-1}) - \left(\frac{1}{\sqrt{2}} \times \frac{\pi}{4} - \sqrt{1-\frac{1}{2}} \right) + \left(\frac{1}{\sqrt{2}} \times \frac{\pi}{4} + \sqrt{1-\frac{1}{2}} \right) - (0 \sin^{-1}(0) + \sqrt{1-0^2})$$

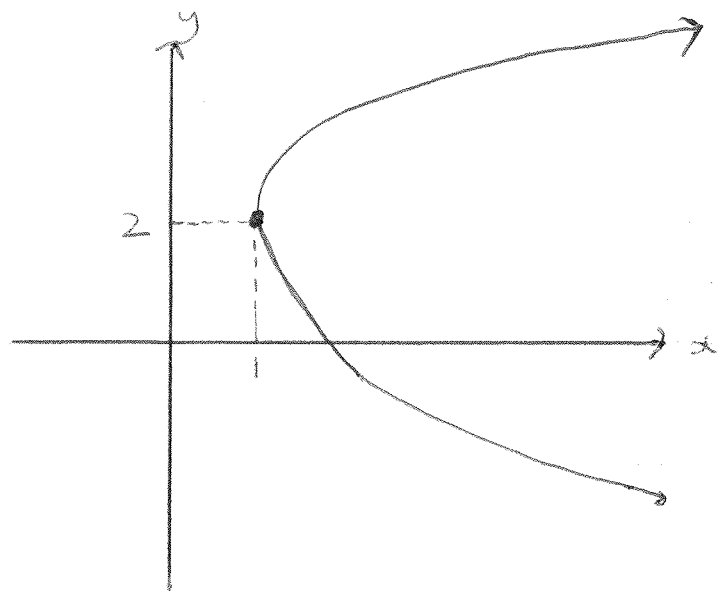
✓

$$= -\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} - 1$$

$$= (\sqrt{2} - 1) \text{ units}^2$$

✓

(c)(i)



$$\begin{aligned} (ii) \quad M &= \left(\frac{(p^2+1) + ((-p)^2+1)}{2}, \frac{2(2p+1) + 2(2(-p)+1)}{2} \right) \\ &= \left(\frac{2p^2+2}{2}, \frac{4p+2-4p+2}{2} \right) \\ &= (p^2+1, 2) \end{aligned}$$

$$y = 2, \quad x > 1$$

this is the axis of the parabola

(iii) equation of line is $y = mx + 5$

substituting in $(t^2+1, 2(2t+1))$

$$2(2t+1) = m(t^2+1) + 5$$

$$4t + 2 = mt^2 + m + 5$$

$$\therefore mt^2 - 4t + m + 3 = 0$$

Marks

$$\Delta > 0$$

$$\therefore (-4)^2 - (4)(m)(m+3) > 0$$

$$16 - 4m^2 - 12m > 0$$

$$m^2 + 3m - 4 < 0$$

$$(m+4)(m-1) < 0$$

$$\therefore -4 < m < 1$$

$$\text{but } m \neq 0$$

✓

✓